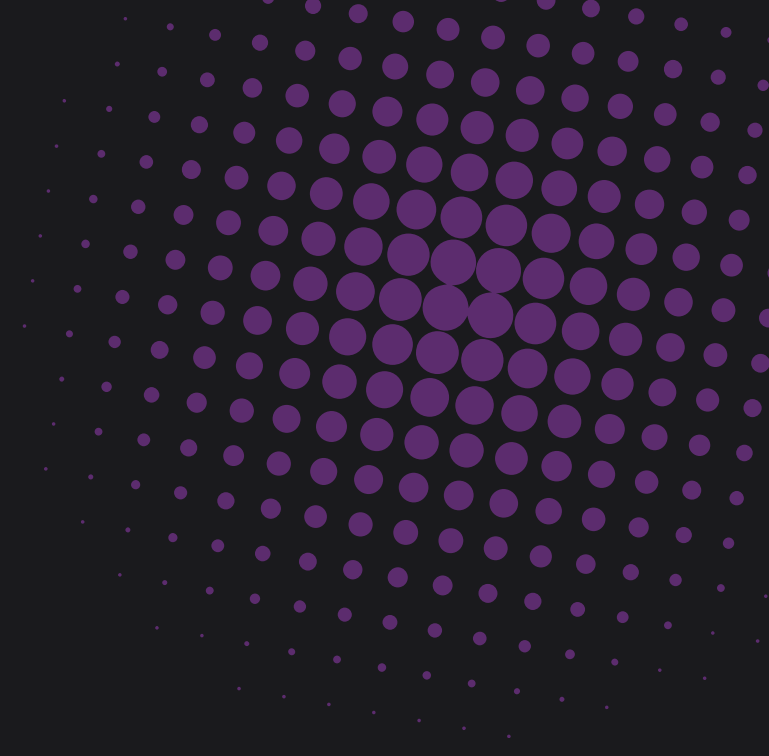
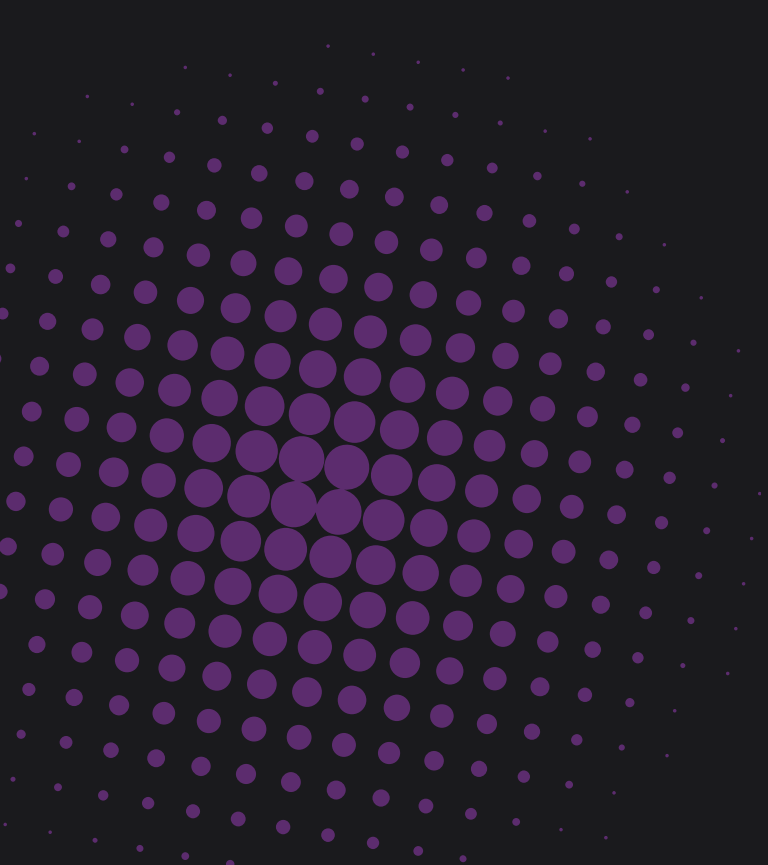




CLIPMIND AI

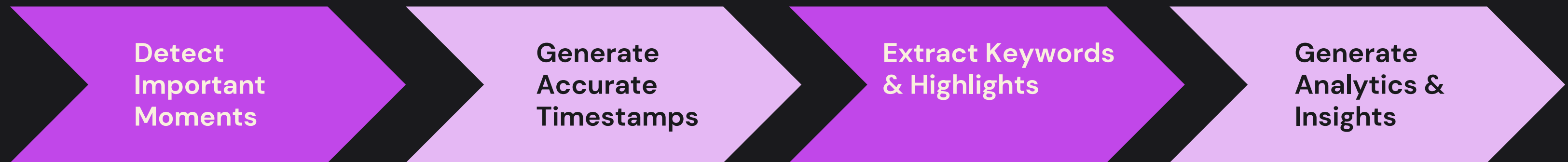
AI Video Summarization & Key
Moment Detection Platform

MILESTONE 3



OBJECTIVE

Develop the AI pipeline that:



Goal

Provide users with quick access to the most important parts of long videos through AI-powered key moment detection and analytics.

Features Implemented

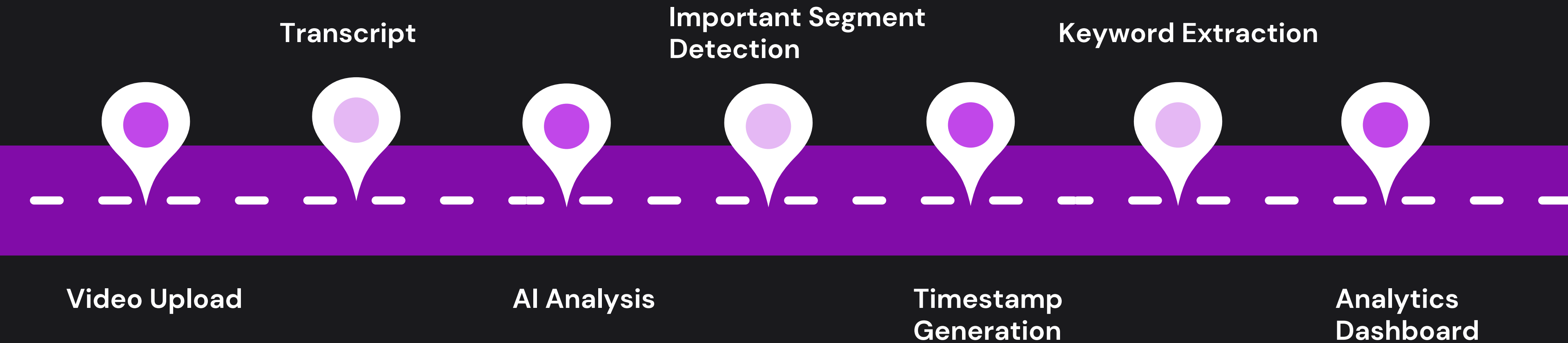
Timestamp Generation
Automatically identifies important timestamps within videos.

Key Moments Detection
Highlights the most relevant sections using transcript analysis.

Keyword Extraction
Extracts significant keywords and topics from the transcript.

Analytics Dashboard
Displays processing statistics, insights, and content analytics.

Workflow



Technology Stack

Frontend

- React.js
- Next.js
- Tailwind CSS

Backend

- FastAPI (Python)
- JWT Authentication

Database

- PostgreSQL
- MongoDB

AI & Machine Learning

- OpenAI Whisper (Speech-to-Text)
- Hugging Face Transformers
- BART / T5
- PyTorch

Video Processing

- FFmpeg
- OpenCV

Deployment & Tools

- Docker
- Git & GitHub
- VS Code
- Postman

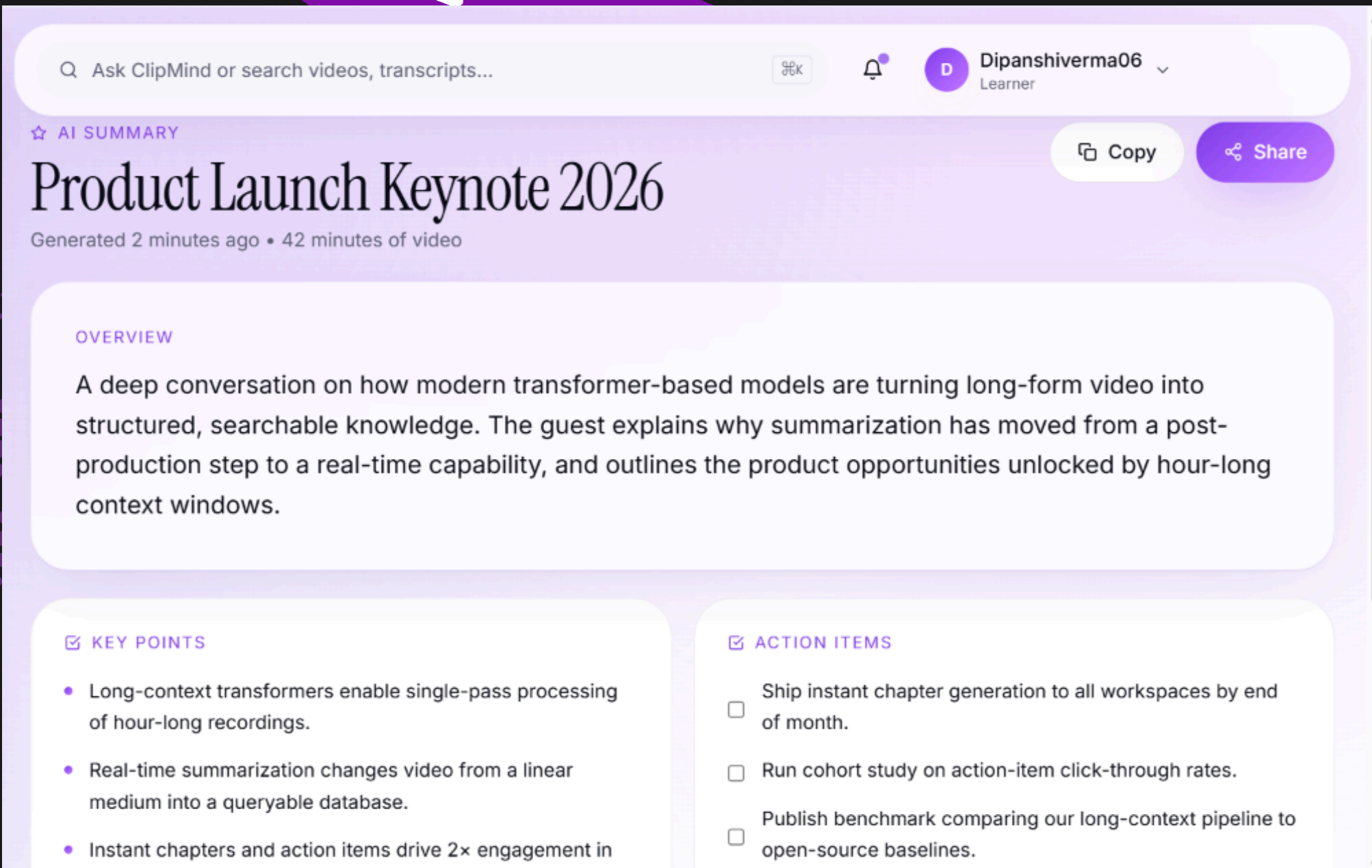
Key Moments Detection

Features Implemented

- Detects important video segments
- Generates accurate timestamps
- Extracts video highlights
- Identifies key discussion points
- Segments content into meaningful topics

Workflow

Video → Transcript → AI Analysis → Key Moments Detection → Timestamp Generation → Highlight Extraction



The screenshot displays the ClipMind AI interface. At the top, there is a search bar with the text "Ask ClipMind or search videos, transcripts...". To the right of the search bar are icons for a document, a bell, and a user profile labeled "Dipanshivarma06 Learner". Below the search bar, the title "Product Launch Keynote 2026" is prominently displayed, with a subtitle "Generated 2 minutes ago • 42 minutes of video". To the right of the title are "Copy" and "Share" buttons. The main content area is divided into three sections: "OVERVIEW", "KEY POINTS", and "ACTION ITEMS". The "OVERVIEW" section contains a paragraph about modern transformer-based models. The "KEY POINTS" section lists three bullet points about long-context transformers, real-time summarization, and instant chapters. The "ACTION ITEMS" section lists three tasks: shipping instant chapter generation, running a cohort study, and publishing a benchmark.

Q Ask ClipMind or search videos, transcripts...

☆ AI SUMMARY

Product Launch Keynote 2026

Generated 2 minutes ago • 42 minutes of video

COPY SHARE

OVERVIEW

A deep conversation on how modern transformer-based models are turning long-form video into structured, searchable knowledge. The guest explains why summarization has moved from a post-production step to a real-time capability, and outlines the product opportunities unlocked by hour-long context windows.

KEY POINTS

- Long-context transformers enable single-pass processing of hour-long recordings.
- Real-time summarization changes video from a linear medium into a queryable database.
- Instant chapters and action items drive 2× engagement in

ACTION ITEMS

- ☐ Ship instant chapter generation to all workspaces by end of month.
- ☐ Run cohort study on action-item click-through rates.
- ☐ Publish benchmark comparing our long-context pipeline to open-source baselines.

Analytics Dashboard

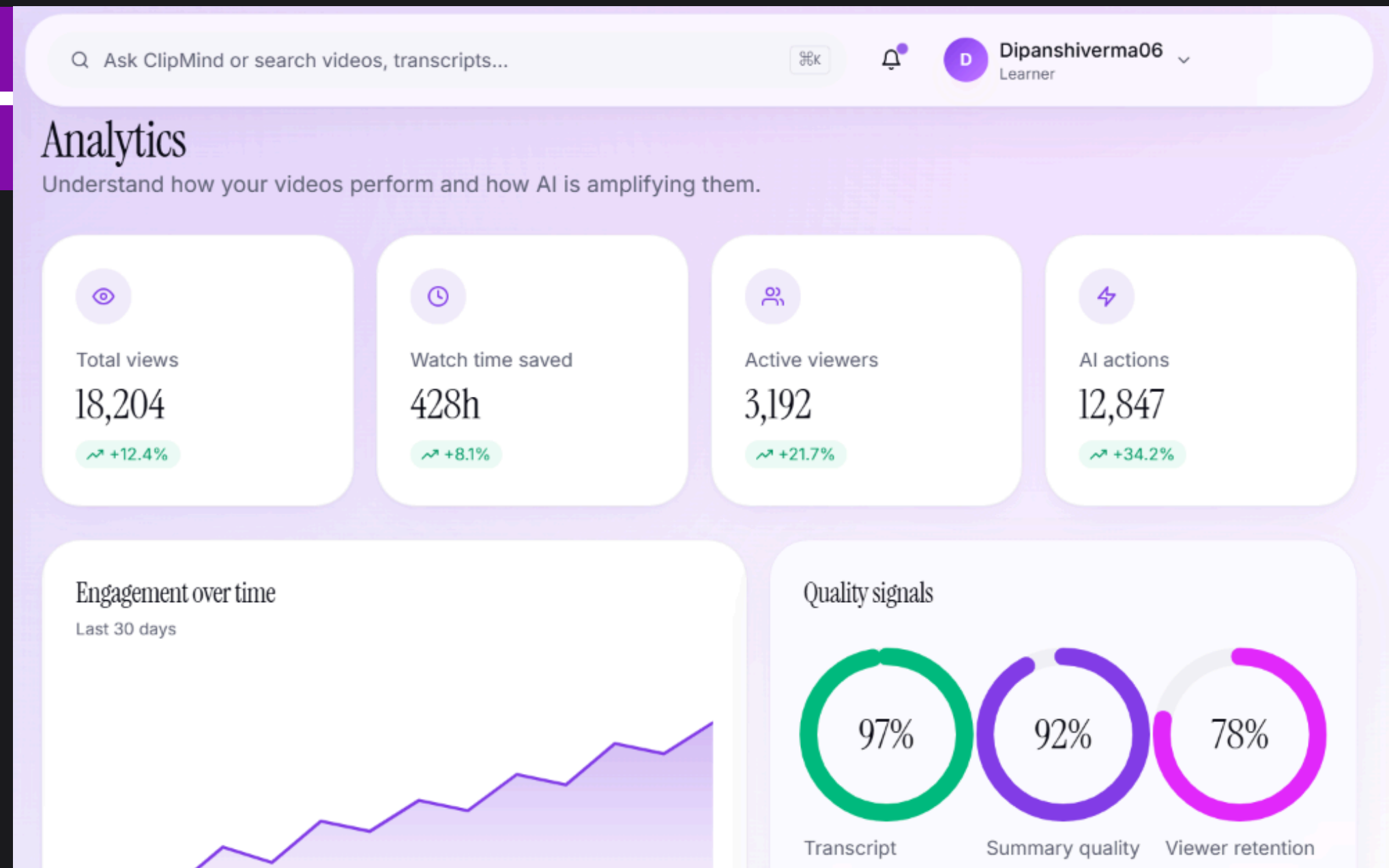
Video statistics

Top keywords

Content insights

Processing reports

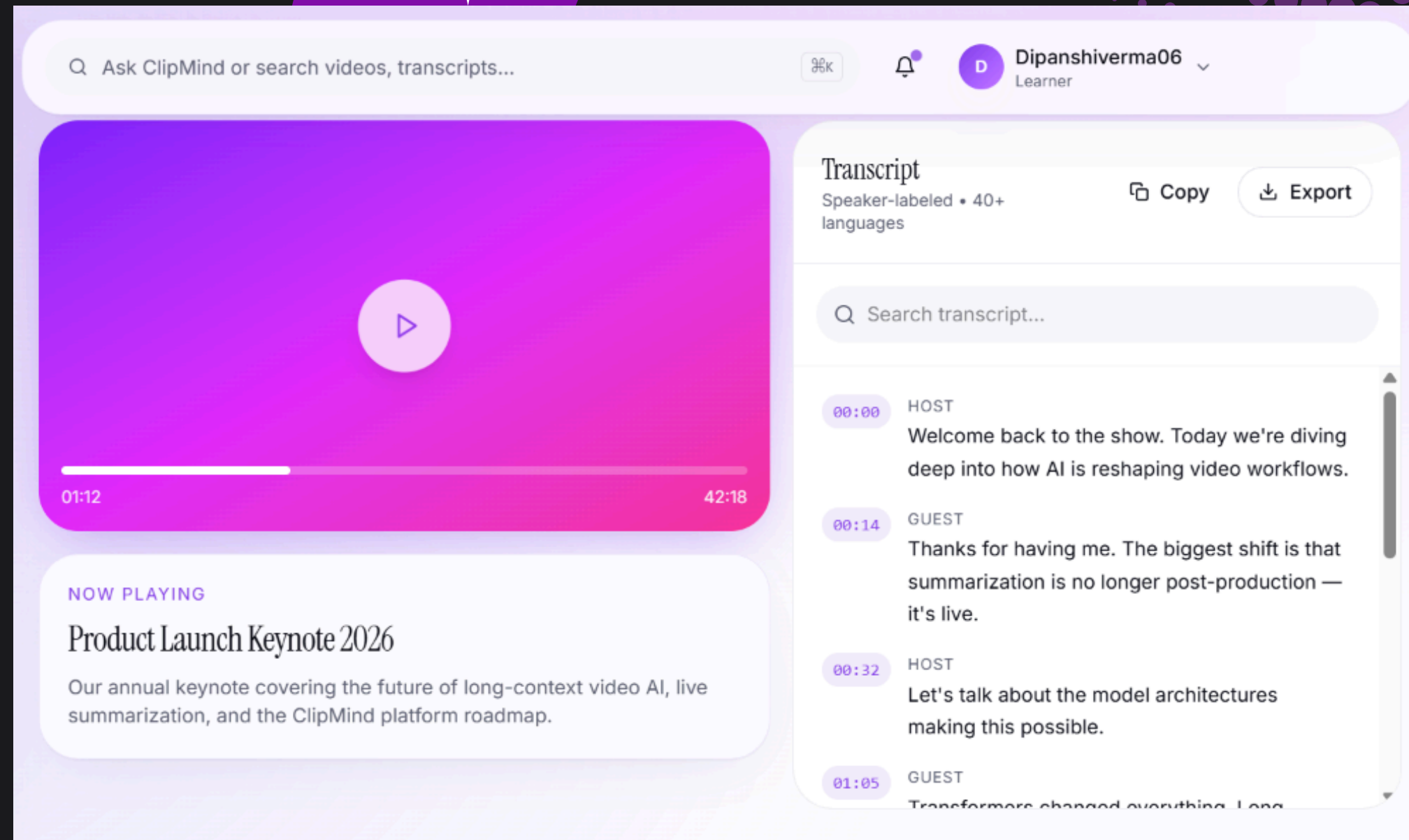
Usage analytics



Results

● Outputs Generated

- Key moments detected successfully
- Accurate timestamps generated
- Relevant highlights extracted
- Important keywords identified
- Analytics dashboard created
- Improved navigation through long videos



Challenges & Solutions

Challenges

- Accurate key moment detection
- Timestamp precision
- Dashboard performance
- Keyword relevance

Solutions

- Combined transcript analysis with AI
- Optimized detection logic
- Efficient API integration
- NLP preprocessing

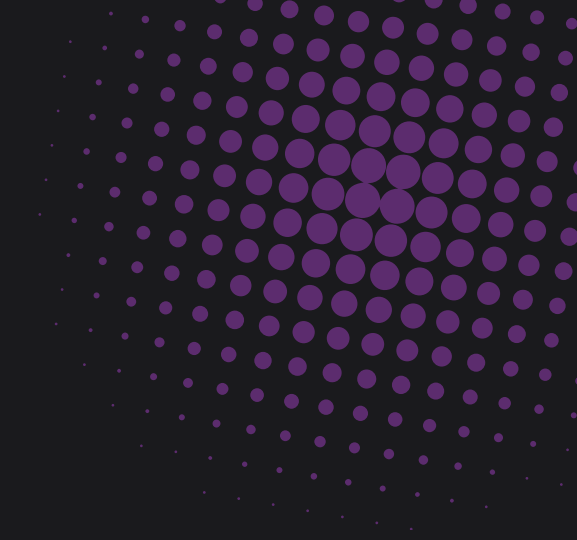
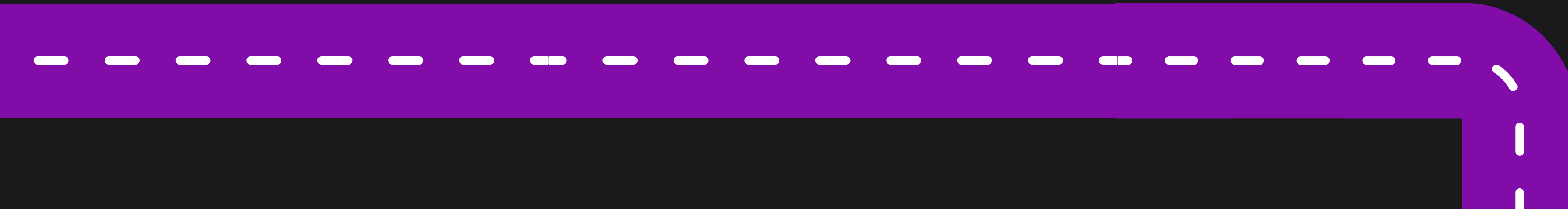
Conclusion & Next Milestone

Milestone 3 Completed

- Key Moments Detection
- Timestamp Generation
- Analytics Dashboard
- Keyword Extraction
- Content Insights

Next Milestone

- Testing
- Docker Deployment
- Cloud Deployment
- Performance Optimization
- Documentation
- Final Demo



THANK YOU

Turning long-form videos into meaningful insights with AI.